TANISHQ TEWARI

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PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate who completed a **2-month in-campus Artificial Intelligence & Machine Learning program at the Indian Institute of Technology (IIT) Mandi**, covering deep learning, convolutional neural networks (CNNs), NLP and model deployment. Team leader of the **2nd-place team at the OFFGRID Hackathon** (IEEE Student Branch, Manipal University Jaipur). Builds and deploys end-to-end AI/ML systems with Python, PyTorch, Scikit-Learn, FastAPI and React, including a graph neural network debugger for LLM agents (93.1% root-cause accuracy) and a multilingual NLP triage platform that cut duplicate tickets by 56%. Seeking Software Engineering and AI/ML internships.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, C, SQL, JavaScript, TypeScript
Machine Learning & AI: Machine Learning, Deep Learning, Natural Language Processing (NLP), Graph Neural Networks (GNN), Large Language Model (LLM) Agents, PyTorch, Scikit-Learn, XGBoost, Data Preprocessing, Feature Engineering, Model Training, Hyperparameter Tuning, Cross-Validation, Model Evaluation, Model Deployment, Optical Character Recognition (OCR)
Backend Development: FastAPI, REST APIs, SQLite, OpenTelemetry, Role-Based Access Control, Authentication
Frontend Development: React, TypeScript, JavaScript, HTML, CSS, Tailwind CSS, Progressive Web Apps (PWA)
Data & Libraries: NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook
Developer Tools & Deployment: Git, GitHub, GitHub Actions (CI/CD), Docker, Linux, Unit Testing (Pytest), VS Code, Render, Ollama, Groq API
Core Computer Science: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks

PROJECTS

GalliGraph: AI Civic Complaint Triage Platform (2nd Place, OFFGRID Hackathon) Oct 2026
Python, Scikit-Learn, NLP, FastAPI, React, TypeScript, Docker | github.com/Tanishqtiwari1/GalliGraph | Live Demo

- Led a 2-member team to build an AI system that converts free-text civic complaints in English, Hindi and Hinglish into a ranked, de-duplicated action queue for municipal officers.
- Trained a TF-IDF (word and character n-gram) + Logistic Regression classifier on 14,491 real complaints (2019–21), reaching 80.6% accuracy on 1,548 unseen 2022 complaints vs. 69.6% for keyword rules (+11 points), and 97% on a synthetic Hindi/Hinglish test set.
- Engineered geo-temporal duplicate detection with 96.4% precision (F1 0.90), reducing tickets by 56% (665 to 292), plus explainable priority scoring, service-level agreement (SLA) escalation and Poisson-based spike detection at ~15 ms per complaint on CPU.
- Shipped citizen and officer portals with role-based access control, scrypt password hashing, signed sessions and rate limiting; containerized with Docker and deployed on Render.

Black Box: Flight Recorder & Debugger for AI Agents (BNB'26 CodeStorm Hackathon, GDG on Campus) Oct 2026
Python, PyTorch, GNN, FastAPI, OpenTelemetry, LLM Agents, Pytest | github.com/Tanishqtiwari1/BlackBox | Live Demo

- Built a debugging platform that records every LLM decision and tool call of an AI agent as OpenTelemetry traces, then localizes the root cause of failures, replays the run and verifies the fix.
- Designed a PyTorch graph neural network with directional message passing over trace graphs, achieving 93.1% top-1 root-cause accuracy (88.4% on unseen fault types) vs. 65.7% for a rule-based baseline, evaluated on 5,000 generated agent runs with injected faults.
- Implemented checkpointed, event-sourced replay that re-executes only 34% of steps; 90.9% of failures fixed by repairing the top-ranked step and 98.1% within 3 replays; covered by 29 Pytest tests and deployed live.

CardioGuard: AI Cardiovascular Risk Screening Platform Sep 2026
Python, XGBoost, Scikit-Learn, FastAPI, React, Progressive Web App | github.com/Tanishqtiwari1/CardioGuard | Live Demo

- Trained an XGBoost classifier with grid-search hyperparameter tuning on 918 real patients (UCI Heart Disease), achieving 82.1% accuracy, 83.9% F1 score and 0.915 ROC-AUC on a held-out test set.
- Detected data leakage in a popular Kaggle dataset where duplicate rows inflated accuracy to 97%, and rebuilt the data preprocessing pipeline on clean, de-duplicated data.
- Built an offline-first Progressive Web App for community health (ASHA) workers with an in-browser Framingham risk model (4,240 adults, 0.73 cross-validated ROC-AUC), Hindi/English UI, OCR of lab reports and PDF report generation; automated deployment with GitHub Actions.

ACHIEVEMENTS

- **2nd Place, OFFGRID Hackathon**, TechIdeate 2026, IEEE Student Branch, Manipal University Jaipur (Oct 2026), as Team Leader of Team Codestorm.
- **Artificial Intelligence & Machine Learning In-Campus Certification**, Indian Institute of Technology (IIT) Mandi (May – Jul 2026): Supervised Learning, Deep Learning, Convolutional Neural Networks (CNNs), NLP and Model Deployment.

EDUCATION

Manipal University Jaipur | B.Tech, Computer Science and Engineering (AI & ML) Oct 2024 – Aug 2028
Relevant Coursework: Data Structures and Algorithms, Advanced Data Structures, Object-Oriented Programming, Machine Learning, Computer Vision, Database Management Systems, Computer Networks, Embedded Systems